

NewsBot Intelligence System 2.0 - **Technical Documentation**

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1) Introduction

Where the prototype is a work-in-progress as of mid-semester, this project concept applies a production-style news analysis engine, synthesizing many NLP components into one uniform, end-to-end pipeline: supervised classification, topic modeling, POS, sentiment, extractive summarization, semantic search, a bare-minimal multilingual foray and a bare-minimal interactive front end. Everything runs in Colab, produces verifiable results to /content/out and requires little-to-no setup for a grader (or teammate) to replicate.

2) Data Pipeline

Ingestion. The code accepts either a ZIP with CSV inside or a direct CSV. It autodetects a reasonable text column name (text, Text, content, body, etc.).

Cleaning. Light normalization (strip extra whitespace, meaning-carrying punctuation, drop duplicates/blanks). A standardized file - `bbc_clean.csv` - is saved to /content/data.

Labels (if missing). If the CSV lacks labels, the notebook generates them via TF-IDF + MiniBatchKMeans. Tiny clusters get merged into misc so the later stratified split doesn't fail. If labels do exist, the system uses them as-is.

3) Supervised Learning

Split. Stratified 80/20 train/test (if each class has ≥ 2 elements.) \ Vectorize. TF-IDF with English stopwords, `ngram_range=(1,2)`, `min_df=2`, `max_df=0.9`, `max_features=50k`. \ Model. Trained Logistic Regression and Linear SVM; whichever is better automatically selected. \ Save diagnostics.

- `metrics.json` (accuracy and by-class metrics)
 - `confusion_matrix.png`
 - `misclassified_samples.csv`

These files provide quick, trustworthy evidence of system performance and where mistakes tend to group.

4) Topic Modeling

Two models suggest hidden topics:\ NMF (on TF-IDF) - typically more interpretable.\ LDA (on CountVectorizer) - takes a probabilistic perspective.\ Exports:

- topics_overview.csv (model, topic id, top terms)
- doc_topics_nmf.csv (topic weights for each document using NMF)

5) POS & Writing Style

Using NLTK's universal tagset, the notebook takes samples of articles per category and combines the count of the tags (NOUN, VERB, ADJ, etc.). This shows differences in style (ex. sports recaps are more verbal/action based; tech/finance are more noun based). \Export: pos_by_category.csv.

6) Sentiment

VADER is utilized for sentence sentiment (pos/neg/neu + compound). It's lightweight, consistent, and works well with headlines and punchy news snippets. Where the multilingual demo is, sentiment is run on translatable English only and stored with it.

7) Extractive Summarization

The TextRank (via sumy) takes the most informative sentences that are necessary, avoiding some hallucination and heavily relied upon use. \ Export: summaries.csv (a sample of summaries).

8) Semantic Search

A TF-IDF cosine is a search that returns the top-k most similar articles to an input query.\ Export: search_examples.csv with queries, scores of relevance, and snippets.

9) Multilingual Intelligence

A practical cross-language approach:

1. Input translation to English (deep_translator).
2. Apply English tools (vader sentiment; classification/search if needed). Export: multilingual_eval.csv with original, translated text, and sentiment scores.

10) Conversational Interface (Interactive)

A practical cross-language approach:

Input translation to English (deep_translator).

Apply English tools (vader sentiment; classification/search if needed). Export: multilingual_eval.csv with original, translated text, and sentiment scores.

11) Reproducible Outputs (for grading & repo)

Everything lands in /content/out:

- metrics.json
- confusion_matrix.png
- misclassified_samples.csv
- topics_overview.csv
- doc_topics_nmf.csv
- summaries.csv
- search_examples.csv
- multilingual_eval.csv
- models/*.joblib (best pipeline + vectorizer)

12) Implementation Choices & Trade-offs

- TF-IDF + linear models: transparent, speedy, and sufficiently precise for categories with balanced news.
- NMF + LDA: they are complementary perspectives of topics; NMF is clearer from TF-IDF, LDA is more probabilistic with bag-of-words.
 - VADER: not fancy but effective for general English and inexpensive.
- Translate-then-analyze: straightforward and useful for a final demonstration; true multilingual models would take more to implement.
 - Gradio: tiny UI, huge potential for demonstration.

13) How to Run (Colab)

1. Run the Setup cell (creates folders, installs light deps).
2. Upload the ZIP/CSV or place a CSV in /content/data.
3. Run the cells in order up until "Final outputs to include."
4. Download /content/out/* and commit to Github.
5. Leave the notebook + /out folder in your repo for grading.

14) Limitations & Next Steps

- Use sentence-BERT embeddings for enhanced retrieval.
- Perform light abstractive summarization (T5-small) on a subset.
- Add NER + entity linking to bring surface entities/people/orgs/places.
 - Monitor topic drift if there are timestamps.

15) Conclusion

This system combines all necessary modules into one, trustworthy pipeline that operates on Colab's free version and generates outputs ready for an audit. It's a stable foundation for something that could eventually be newsroom quality.